VBF-T2R1NOFORCE-BOM-001 - Bill of Materials (F1)

Sheet: BOM

Component	Mass (g/cell)	kg/kWh	Source / note
Positive electrode active material (NMC811)	16.842	0.9436	layer_kg_m2.positive_electrode 0.16399 kg/m2 x 0.1027 m2 (calc-energy, electrolyte excluded)
Negative electrode active material (graphite, ceramic-coated)	8.409	0.4711	layer_kg_m2.negative_electrode 0.08188 kg/m2 x 0.1027 m2; coating = ALD Al2O3/TiO2 class, k_SEI 4e-13
Positive current collector (Al, 16 um)	4.437	0.2486	layer_kg_m2.positive_cc x area; density 2700 kg/m3 (parameter set)
Negative current collector (Cu, 12 um)	11.042	0.6187	layer_kg_m2.negative_cc x area; density 8960 kg/m3 (parameter set)
Separator (12 um, porosity 0.47)	0.259	0.0145	layer_kg_m2.separator x area; density 397 kg/m3 (parameter set)
Electrolyte (EC/EMC + LiPF6)	8.226	0.4609	pore volume 6.855 cm3 x 1.2 g/cm3 (literature density, annotated; not in parameter set)
Positive conductive additive / binder	Not modeled	Not modeled	parameter set has no binder/additive volume (volfrac 0.665 + porosity 0.335 = 1.0)
Negative conductive additive / binder	Not modeled	Not modeled	same as positive (volfrac 0.75 + porosity 0.42 = 1.17 > 1: porosity design change; note)
Enclosure / tabs	Not modeled	Not modeled	beyond pure-simulation boundary
TOTAL (electrolyte excluded, contract caliber)	40.99	2.2965	calc-energy mass_kg 0.0409897 kg; energy 17.8485 Wh
TOTAL (electrolyte included, annotated)	49.216	2.7574	sum of rows above; energy 17.8485 Wh (r5_f1_energy.json)